Note that A and B are incorrect because the time-weighted rate of return (TWR) of the fund is the same as the geometric mean return of the fund and is positive:

$$R_{TW} = \sqrt[3]{(1.15)(1.14)(0.96)} - 1 = 7.97\%.$$

7. A is correct. Computation of the money-weighted return, r, requires finding the discount rate that sums the present value of cash flows to zero. Because most of the investment came during Year 2, the money-weighted return will be biased toward the performance of Year 2 when the return was lower. The cash flows are as follows:

   $CF_0 = -10$

   $CF_1 = -100$

   $CF_2 = +120.31$

   The terminal value is determined by summing the investment returns for each period [(10 × 1.14 × 1.08) + (100 × 1.08)].

   $$\frac{CF_0}{(1+IRR)^0} + \frac{CF_1}{(1+IRR)^1} + \frac{CF_2}{(1+IRR)^2}$$
   $$\frac{-10}{1} + \frac{-100}{(1+IRR)^1} + \frac{120.31}{(1+IRR)^2} = 0 .$$

   This results in a value of IRR = 8.53 percent.

   The time-weighted return of the fund is calculated as follows:

   $$R_{TW} = \sqrt[2]{(1.14)(1.08)} - 1 = 10.96\%.$$

8. B is correct. The annualized rate of return for

   ETF 1 annualized return = $(1.0425^{365/125}) - 1 = 12.92\%$

   ETF 2 annualized return = $(1.0195^{52/8}) - 1 = 13.37\%$

   ETF 3 annualized return = $(1.1718^{12/16}) - 1 = 12.63\%$

   Despite having the lowest value for the periodic rate, ETF 2 has the highest annualized rate of return because of the reinvestment rate assumption and the compounding of the periodic rate.

9. A is correct. The continuously compounded return from $t = 0$ to $t = 1$ is $r_{0,1} = \ln(S_1/S_0) = \ln(186.75/208.25) = -0.10897 = -10.90\%$.

10. A is correct.

    $$R_L = R_p + \frac{V_B}{V_E}(R_p - r_D)$$

    $$= 10\% + \frac{USD5\ million}{USD20\ million}(10\% - 6\%)$$

    $$= 10\% + 0.25 \times 4\% = 11.0\%.$$

11. C is correct. Gross returns are calculated on a pre-tax basis; trading expenses *are* accounted for in the computation of gross returns as they contribute directly to the returns earned by the manager. A is incorrect because investment managers' gross returns are pre-tax and not adjusted for risk. B is incorrect because managers' gross returns do reflect the deduction of trading expenses since they contribute directly to the return earned by the manager.